TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Vision-enhanced AI-based control for object capture (gripper)**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	Industrial/ General Machinery	Other Industrial/General Machinery Technologies
Concept of Proposed Technology	The proposed technology consists in using cameras observing a robotic gripper as sensory output to close a feedback loop based on AI-based control methodologies. Sensor placement in thin robotic structures can be challenging. Vision-enhanced control may enable closed-loop solutions without requiring sensor integration but by using cameras external to the actual robotic structure.			
Rationale for Joint R&D Project	- Definition of concrete use case provided by SME - SME can provide hardware components / robotic gripper - SME can enable increase in TRL with respect to current state-of-the-art			
Key Words	1. Robot control using vision as feedback 2. Automated control loop design/learning through visual feedback 3. Path planning / task execution of robotic arm using visual inputs			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5
R&D Objectives	The main R&D objectives are to apply diverse Artificial Intelligence (AI) techniques such as Machine Learning (ML) to robotic manipulators. Using AI shall permit the use of a video stream as control feedback. There are two stages where we see this technology. A low-level use case where the video flux could replace sensors inside manipulators, with such technology the complexity of actuator design is reduced. It should also reduce time spent on designing controllers as the latter can be automated through ML or other similar algorithms. At a higher level, tasks such as pick and place or assembly job could use imitation learning so that less programming is needed.			
R&D Target Specifications	The performance will be evaluated in different terms. On the low-level side, it will be the robotic manipulator placement accuracy, typical values will strongly depend on hardware (camera resolution, optics used, manipulator drive technology etc..). A target performance is thus hard to quantify at this stage and will be defined with the SME. Performance will also include other metrics like robustness to environment and manipulator parameters. Stability and settling time will also be key performance indicators. At task level, the main figure will be the success rate of the task again it shall be discussed with the involved SME but success rate as high as 99.99% is usually required.			
R&D Plan	The plan is to start simple and control a single axis with a video stream. The tasks are the following: 1) Create data collection to be used by AI algorithm, typically video recordings while the axis is driven with PRBS signal. 2) Choose relevant neural network architecture and train it. 3) Test the obtained network against real placement tasks iterate to task 2 if performance is not met. 4) Increase the number of axis and cameras and repeat tasks 1 to 3. 5) Switch to task level. 6) Do a new data collection with several tasks to be learnt. 7) Choose relevant neural network architecture and train it. 8) Test the obtained network against real tasks and iterate to task 7 if performance is not met.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM will provide control algorithm development expertise and knowledge in AI. CSEM will develop all AI software including data collection, training and tests. If needed and agreed with the SME, CSEM can develop the firmware tools needed to interface with the hardware.
Expected R&R of Korean SME Partner	As end user, the SME will define the use cases, contribute to the software integration in a real-world scenario and be responsible for final performance tests. SME will provide all the hardware needed (except computers needed for AI).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The main outcome of the project will be proof that manipulators could be controlled through a video stream. Experimental algorithms will be developed that permits this control on selected hardware. A second optional outcome is task learning, with additional algorithm for task execution.

6. References

(1) https://indico.esa.int/event/321/contributions/6341/

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization